

BENXIANG XIAO

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EDUCATION

Southern University of Science and Technology, Shenzhen, Guangdong, China 2023 – Present

- B.S. in Data Science (Junior Year)
- **Cumulative GPA:** 3.45/4.0 (Rank: 58/74); **Major GPA:** 3.56/4.0 (Rank: 47/74)
- **Courses:** Discrete Mathematics, Probability Theory and Mathematical Statistics, Data Structures and Algorithms, Operations Research and Optimization, Advanced Natural Language Processing, Distributed Storage and Parallel Computing
- **Research Interests:** ML Systems, Efficient LLM Inference, Distributed Systems, HPC

PUBLICATION

[1] Jiahao Fan, Zongsheng Li, **Benxiang Xiao**, Qingzhu Zhang, Xinke Shen, and Quanying Liu. "Semantic-guided Contrastive Learning for EEG Decoding of Video Semantics". In *Proceedings of the 2025 China Automation Congress (CAC)*, IEEE, 2025. doi: 10.1109/CAC67268.2025.11487820.

SKILLS

- Programming Skills: Python, C/C++, Java, Shell, CUDA, SIMD, OpenMP
- Frameworks: PyTorch, TensorFlow, PyTorch Lightning, MindSpore
- Profilers: Kunpeng DevKit, VTune, PyTorch Profiler, IPM, HipProf, MindSpore Profiler, Nsight Systems
- Languages: English, Chinese (native)

RESEARCH & ENGINEERING EXPERIENCE

GPU Sharing | Research Intern Apr. 2026 – Present

With the group of Prof. Dingwen Tao, Institute of Computing Technology, Chinese Academy of Sciences (ICT)

- Conducting research on GPU sharing and resource management for deep learning inference workloads in multi-tenant environments.
- Investigating systems-level bottlenecks in shared-GPU inference scenarios to guide the design of more efficient workload placement and scheduling strategies.

Efficient Inference for Large Models | Research Intern Oct. 2025 – Present

With the group of Prof. Tianwei Zhang, Nanyang Technological University (NTU)

- Investigated systems-oriented approaches for accelerating dynamic inference workloads while preserving model quality.

Distributed Training Theory | Research Intern Jun. 2025 – Feb. 2026

With the group of Prof. Zhi Wang, Tsinghua Shenzhen International Graduate School (Tsinghua SIGS)

- Studied theoretical aspects of efficient distributed training in cloud-edge-end architectures, including derivations for optimization behavior and convergence-related analysis.
- Explored the rationality of Adam convergence.

High-Frequency Financial Data Distributed Computing | Developer Sep. 2025 – Dec. 2025

Course Project, Ranked Top 1% in Performance Benchmark

- Built a high-concurrency quantitative factor calculation system for limit order book data on **HDFS + MapReduce**, optimizing key compression, shuffle, object reuse, and custom CSV parsing to reduce network I/O and GC overhead.
- Achieved a **13× speedup** over baseline with near-C/C++-level efficiency in course evaluation while preserving numerical precision. [Distributed-Storage-and-Parallel-Computing]

2025 APAC HPC-AI Competition | Core Member Jun. 2025 – Nov. 2025

- Optimized **NWChem** through compilation refactoring and communication tuning; profiled **DeepSeek-R1 (671B)** on multi-node GPU clusters with **Nsight Systems** and **SGLang Profiler** to identify scaling bottlenecks.
- Achieved a **4.82× speedup** for NWChem on a single node and improved DeepSeek-R1 throughout by about **70%** through TP/DP/PP tuning and backend benchmarking. [APAC_HPC-AI]

MindSpore Model Development Competition | Team Leader Oct. 2025 – Nov. 2025

- Profiled **DeepSeek-MoE-16B** and **Qwen1.5-MoE** using **MindSpore Profiler**; optimized expert routing, sparse GEMM, and memory fragmentation bottlenecks, and implemented a fused **Softmax+TopK** operator.
- Designed a dual-path inference engine for **Prefill** and **Decode**, reducing decoding latency by about **80%** and improving end-to-end MoE inference throughput. [MindSpore]

2nd SUSTech SuperComputing Competition | Team Leader Jul. 2025 – Aug. 2025

- Implemented a high-performance **HGEMM** kernel using **WMMA** on Tensor Cores, with size-specific tiling, **double buffering**, **asynchronous caching**, register swizzling, and loop unrolling.
- Achieved performance competitive with or exceeding **cuBLAS** in selected small-to-medium matrix settings while maintaining numerical accuracy. [HGEMM]

9th Huawei ICT Competition Chinese National Finals | Team Leader Feb. 2025 – Mar. 2025

- Profiled the **Hypre** library with **Kunpeng DevKit**, optimized sparse solver kernels with **SVE intrinsics**, and improved **PCG** convergence behavior within the **BoomerAMG** framework.
- Improved computational efficiency by about **30%** for large-scale sparse matrix solving on the Kunpeng platform.

SELECTED ACADEMIC ENGAGEMENTS

SupercomputingAsia (SCA) Mar. 2025 & Jan. 2026

Participated in the Student Cluster Competition Workshop at SCA, which broadened my exposure to HPC systems, large-scale computing, and performance engineering.

Huawei Kunpeng Technical Exchange Activities Dec. 2024

Attended Huawei Kunpeng technical exchange activities related to ARM-based HPC software optimization and performance tuning.

LEADERSHIP & COMMUNITY

SuperComputing Club Leader, SUSTech Jun. 2025 – Present

Organized training and project collaboration for students interested in AI systems and HPC, covering CUDA, profiling, and competition-oriented system optimization.

HONORS AND AWARDS

Honorable Mention, 2026 Mathematical Contest in Modeling	May 2026
Second Class Prize, 2026 ASC Student Supercomputer Challenge	Feb. 2026
3rd Place (Onsite), 8th Place (Overall), IndySCC25	Nov. 2025
Achievement Prize, ISC25 Student Cluster Competition	Jun. 2025
Second Class Prize; Excellent AI Performance, 2025 APAC HPC-AI	Nov. 2025
Third Class Prize, 9th Huawei ICT Competition Chinese National Finals	Mar. 2025
Achievement Prize, MindSpore Model Development Competition	Nov. 2025
Bronze Prize, MindSpore Model Fine-tuning Track	Aug. 2024
Bronze Prize, 2024 Kunpeng Application Innovation Competition	Dec. 2024
First Prize, SUSTech SuperComputing Competition	Aug. 2025
Campus Academic Star, Southern University of Science and Technology	Sep. 2024
Outstanding Student, Southern University of Science and Technology	Sep. 2025